

Project 02 — AI Music & Audio Analyzer

Updated Test & Development Report — v1.1

Item	Current Value
Project Status	Steps 1–18C validated
Documentation Version	v1.1
Platform	Windows Laptop
Python	3.14.7
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Music Input	test_music.mp3
DSP Milestone	Steps 1–17 complete
AI Milestone	Steps 18A–18C complete
Audio Health Score	96 / 100 — Excellent
Dashboard	AI Music Analysis Dashboard generated

1. v1.1 Update Summary

This update records the successful completion of Step 18C — AI Music Report & Dashboard Integration. The project now combines measured DSP features, the project-specific Audio Health Score, musical analysis and Gemini AI interpretation into one professional dashboard.

Milestone	Status
Step 18A — Music Feature Package	PASS
Step 18B — AI Music Interpretation	PASS
Step 18C — AI Report & Dashboard Integration	PASS
Dashboard	reports\ai_music_analysis_dashboard.png

2. End-to-End AI Music Pipeline

test_music.mp3 → DSP Analysis → music_feature_summary.json → Gemini AI → music_ai_interpretation.json / .txt → Step 18C Dashboard → ai_music_analysis_dashboard.png

3. Step 18C Dashboard Contents

Section	Integrated Information
Track Information	File, duration, sample rate, channels
Audio Health	96/100 — Excellent
Rhythm	109.96 BPM, Moderate, beats and onsets
Signal	RMS, peak and dynamic range
Clipping	0 samples, 0.000000%
Frequency	Bass 89.42%, Mid 10.50%, Treble 0.08%
Spectral	Centroid, bandwidth, rolloff, flatness, ZCR
Musical Key	G Minor, correlation 0.7472
AI Interpretation	Overall interpretation and key findings

4. Verified Music Track Results

Parameter	Result
Track	test_music.mp3
Duration	240.02 sec
Sample Rate	44,100 Hz
Channels	1 (mono)
RMS Level	-17.05 dBFS
Peak Level	-0.17 dBFS
Dynamic Range	16.88 dB
Bass / Mid / Treble	89.42% / 10.50% / 0.08%
Dominant Band	Bass
Estimated BPM	109.96
Tempo Category	Moderate
Estimated Key	G Minor
Key Correlation	0.7472
Clipping	0 samples
Health Score	96/100 — Excellent

5. AI Interpretation Outputs

File	Purpose
reports\music_feature_summary.json	Measured feature package used as AI input
reports\music_ai_interpretation.json	Structured Gemini interpretation
reports\music_ai_interpretation.txt	Human-readable AI report
reports\ai_music_analysis_dashboard.png	Integrated Step 18C dashboard

The Gemini response was validated as JSON and stayed within the supplied measurements. It avoided unsupported claims about artist, genre, instruments, vocals and lyrics, and treated the health score as an internal project heuristic.

6. Dashboard Validation

Validation	Status
Feature summary loaded	PASS
AI interpretation loaded	PASS
DSP results integrated	PASS
Health score integrated	PASS
Key and BPM integrated	PASS
Frequency distribution integrated	PASS
Spectral features integrated	PASS
Clipping analysis integrated	PASS

Validation	Status
AI findings integrated	PASS
PNG dashboard generated	PASS

A non-fatal Matplotlib warning occurred because DejaVu Sans does not contain the robot-face emoji used in the dashboard heading. The dashboard was generated successfully. The emoji can be removed for a clean technical run.

7. Completed Project Steps

Step	Feature	Status
1	Audio Device Detection	PASS
2	Airdopes Microphone Verification	PASS
3	Airdopes WAV Recording	PASS
4	WAV Audio Analysis	PASS
5	Waveform Visualization	PASS
6	Audio Level Analysis	PASS
7	FFT / Frequency Spectrum	PASS
8	STFT / Spectrogram	PASS
9	Spectral Feature Analysis	PASS
10	Bass / Mid / Treble Energy	PASS
11	Tempo / Beat Analysis	PASS
12	Integrated Audio Feature Summary	PASS
13	Real Music File Input & Analysis	PASS
14	Professional Music Visualization Dashboard	PASS
15	Key / Chroma Analysis	PASS
16	Loudness / Dynamic Range / Clipping	PASS
17	Audio Quality & Health Score	PASS
18A	Music Feature Package	PASS
18B	AI Music Interpretation	PASS
18C	AI Report & Dashboard Integration	PASS

8. Current Project Structure

```
Project-02-AI-Music-Audio-Analyzer/  
■■■ audio/ → airdopes_test.wav  
■■■ music/ → test_music.mp3  
■■■ reports/ → analysis graphs, dashboards, JSON and AI reports  
■■■ src/ → Steps 1–18C Python scripts  
■■■ tests/ ■■■ venv/ ■■■ requirements.txt ■■■ .gitignore ■■■ README.md
```

9. Security

Gemini credentials are supplied through GEMINI_API_KEY rather than hard-coded into source code. API secrets and local environment files must remain excluded from GitHub.

10. Engineering Limitations

DSP results depend on selected algorithms, sample rate, mono conversion and frequency-band definitions. Key detection is an estimate based on chroma/profile correlation. The Audio Health Score is project-specific. Gemini output is an AI interpretation of measured features and is not a replacement for professional audio engineering or mastering analysis.

11. Version History

- v0.1 — Project-02 documentation baseline created from the beginning.
- v0.2 — Steps 4–6: WAV analysis, waveform visualization and audio-level analysis.
- v0.3 — Steps 7–8: FFT spectrum and STFT spectrogram.
- v0.4 — Step 9: spectral feature analysis.
- v0.5 — Steps 10–11: Bass/Mid/Treble and Tempo/Beat analysis.
- v0.6 — Steps 12–13: integrated feature summary and real music track analysis.
- v0.7 — Step 14: professional music visualization dashboard.
- v0.8 — Steps 15–16: key/chroma plus loudness, dynamic range and clipping analysis.
- v0.9 — Step 17: Audio Quality & Health Score; 96/100 — Excellent.
- v1.0 — Steps 18A–18B: AI-ready feature package and Gemini AI Music Interpretation.
- v1.1 — Step 18C: AI Music Report & Dashboard Integration successfully validated.

12. Next Development Step

Step 19 — AI Music Recommendation & Insight Engine: transform measured features and AI interpretation into actionable recommendations and deeper music insights.